

SMART INDIA HACKATHON 2026



iTantra: Voice That Travels as Meaning

Problem Statement ID – SIH26173

Problem Statement Title – iTantra: Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links

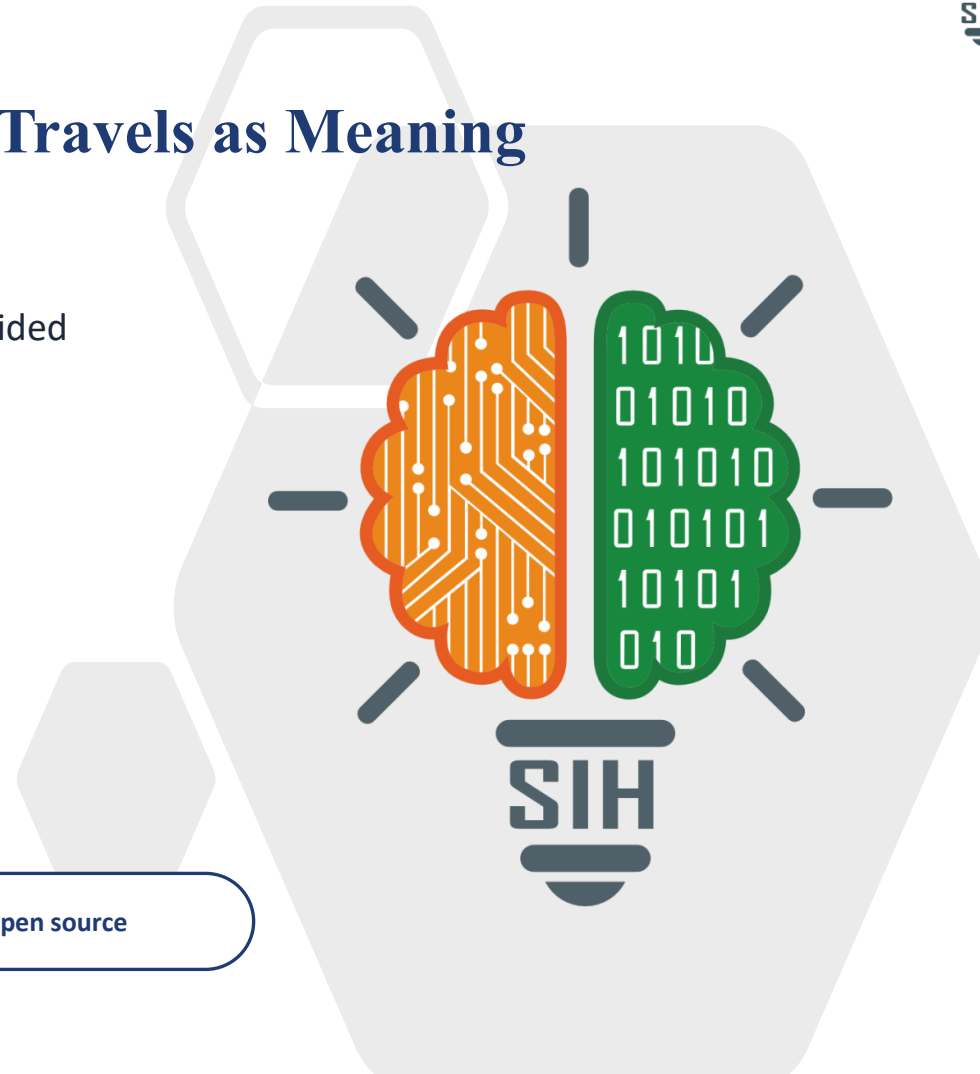
Organisation – Indian Space Research Organisation (ISRO)

Theme – Miscellaneous

PS Category – Software

Team ID –

Team Name (Registered on portal) – Hexabits



Working prototype



10 languages, offline



Fully open source



iTantra Offline speech-to-speech radio for disasters • 10 Indian languages • ~234 bytes per spoken sentence • runs on 3 GB Android phones

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

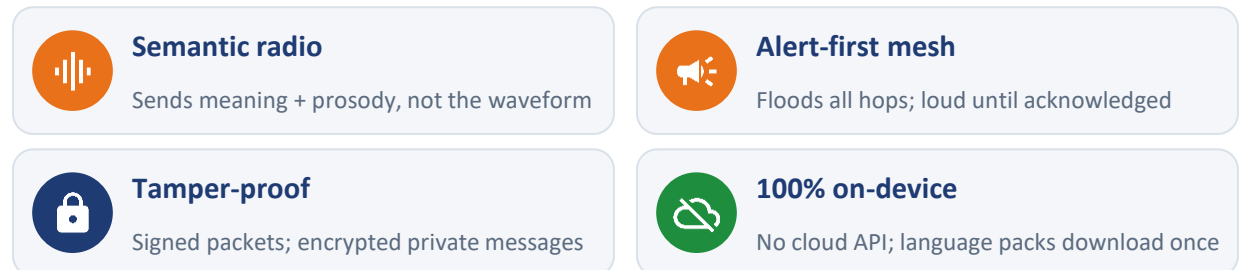
An Android walkie-talkie that turns speech into a few hundred bytes of signed text, relays it phone to phone, and speaks it again on the other side.



✓ How it addresses the problem

- ✓ **Low bitrate:** one sentence is ≈ 234 B (≈ 0.5 kbps) instead of > 64 kbps streamed audio
- ✓ **No network:** phones relay for each other; no tower, internet, SIM or server
- ✓ **Inclusive:** people who cannot read still hear the message, in their own language
- ✓ **Low-end ready:** int8 models, one language in RAM, runs on 3 GB Android phones

🚀 Innovation and uniqueness



≈ 234 B
per spoken sentence

$30 \times$ smaller
than Opus 16 kbps

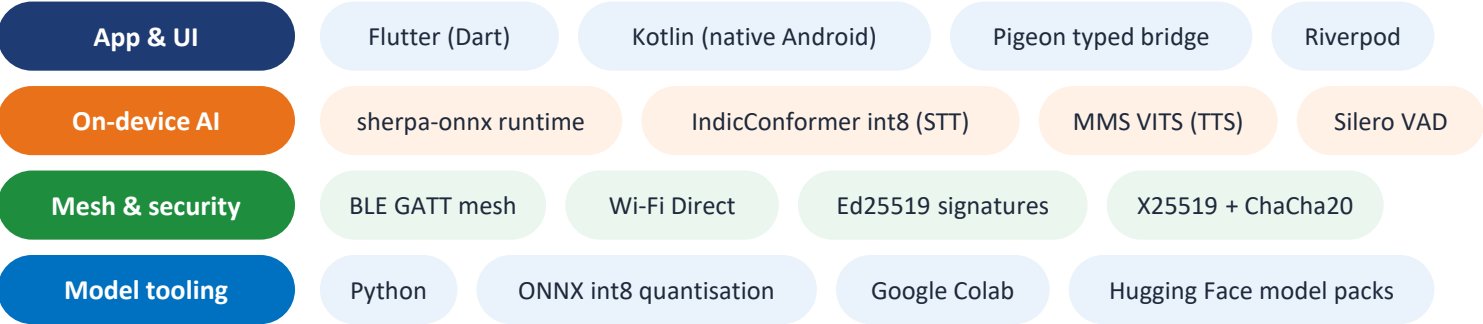
$120 \times$ smaller
than 64 kbps voice



TECHNICAL APPROACH

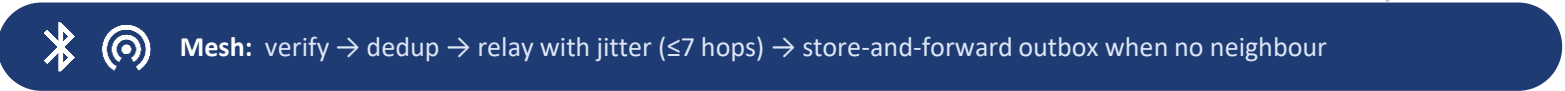
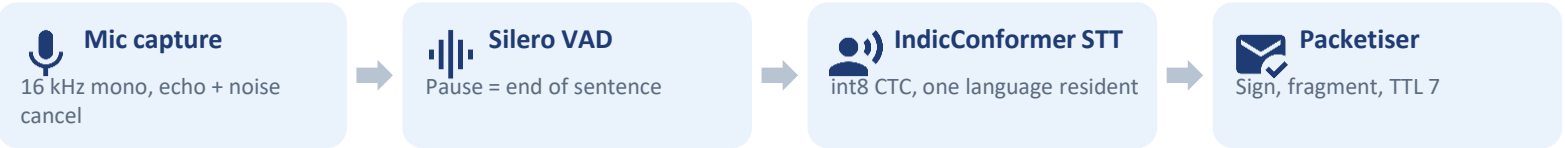


< > Technologies to be used

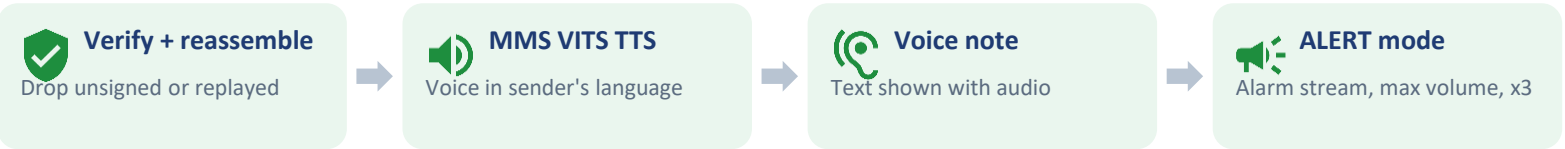


Methodology and process for implementation (flow)

PHONE A • SENDER



PHONE B • RECEIVER



Working prototype (Android)



236 ms
Hindi sentence decode
644 ms
STT model load
654 ms
TTS voice load
10 / 10
languages verified

Talk screen of the running app (redrawn); timings measured on a OnePlus CPH2717, Android 16.

Analysis of the feasibility of the idea

10 / 10

languages running fully offline

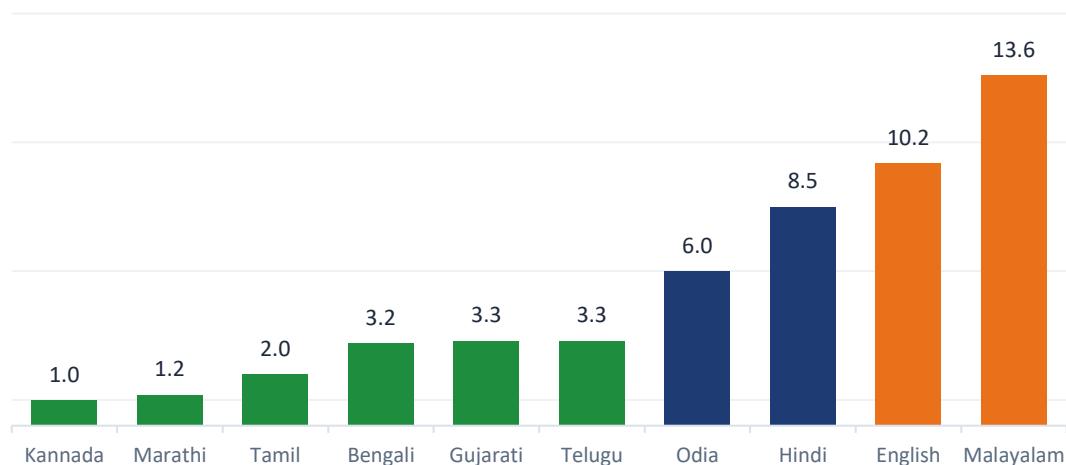
~0.3 GB

per language (speech + voice packs)

<1.1 GB

peak RAM (est.), fits 3 GB phones

Round-trip check: character error % by language (lower is better)



Each language: our TTS voice speaks a flood warning, our STT transcribes it back (sherpa-onnx, CPU, 2 threads). STT ran 5–20 × faster than real time. Next: real speech from 10 speakers per language.

Potential challenges and risks → Strategies for overcoming them

Short range per hop

~60–100 m per Bluetooth hop



7-hop relay, BLE Coded PHY where supported, store-and-forward by moving phones

Hard speech

Noise, soft voices, Malayalam / Odia accuracy



Gain control + tuned VAD, distress hotword list, text shown with voice, real-speech test set

Low-end phones

RAM, battery drain, background kills



int8 models, one language in RAM, duty-cycled scanning, foreground service

Fake alerts and misuse

Spoofing; mesh apps were flagged in India (Jul 2026)



Ed25519-signed packets, verified NDMA / ISRO key, built as an authenticated alert tool

Voice licence

MMS-TTS voices are non-commercial



Swap to CC-BY voices trained on SYSPIN / IndicVoices-R before deployment

85%

of Indians speak one of
our 10 languages as
mother tongue (Census
2011)

~26%

of Indians aged 7+
cannot read; voice
still reaches them
(Census 2011)

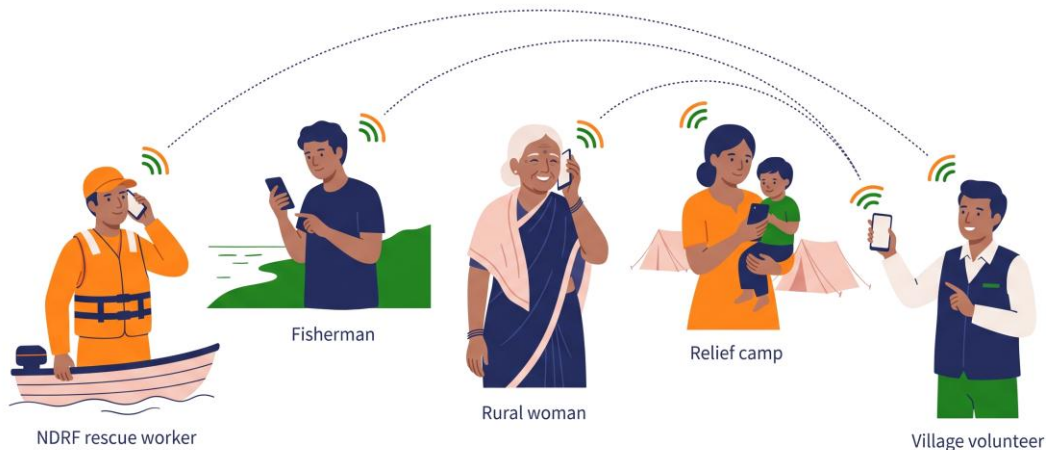
0

towers, internet, SIM cards or
servers needed

₹0

data cost per message, on
phones people already own

Potential impact on the target audience



Benefits of the solution



Social

Voice in 10 languages for non-literate, elderly and visually impaired people; no typing needed.



Economic

Runs on existing low-cost phones; zero infrastructure, data or SMS cost; open models.



Environmental

No diesel towers or generators; low-power Bluetooth saves battery in long outages.



Strategic (ISRO)

Byte-sized frames suit NavIC messaging and LoRa links; data never leaves Indian devices.



Next: "heard by N phones" receipts • SMS gateway when any phone regains signal • voice-preserving prosody • 10-phone field trial

Details / Links of the reference and research work



Models and tools

AI4Bharat IndicConformer ASR (MIT)
github.com/AI4Bharat/IndicConformerASR

sherpa-onnx: offline speech runtime
github.com/k2-fsa/sherpa-onnx

Meta MMS-TTS voices (VITS)
huggingface.co/facebook/mms-tts

Silero VAD
github.com/snakers4/silero-vad

IndicConformer sherpa-onnx int8 exports
huggingface.co/.../indicconformer-sherpa-onnx



Research

IndicVoices: multilingual Indian speech (ACL 2024)
arxiv.org/abs/2403.01926

STCTS: speech at 80 bps via text + prosody
arxiv.org/abs/2512.00451

IndicVoices-R: Indian TTS corpus (NeurIPS 2024)
github.com/AI4Bharat/IndicVoices-R

BitChat mesh protocol whitepaper
github.com/permissionlesstech/bitchat

SYSPIN: open CC-BY Indian TTS dataset (IISc)
vaani.iisc.ac.in/dataset/syspindataset



Real-world evidence

FireChat mesh, Hong Kong 2014 (500k installs)
en.wikipedia.org/wiki/FireChat

BitChat in Nepal 2025: 48,781 installs in a day
forbes.com (Sep 11, 2025)

Census of India 2011: languages, literacy
censusindia.gov.in

NDMA: disaster management in India
ndma.gov.in

Bridgefy offline mesh adds E2E encryption (2020)
techcrunch.com (Nov 2, 2020)



Our work is open source

Android app, mesh protocol spec and model tooling, plus 20 ready language packs
(10 STT + 10 TTS) verified end to end.



App source code
[github.com/nileshpatil6/
SIH-Hexabits](https://github.com/nileshpatil6/SIH-Hexabits)



Offline model packs
[huggingface.co/datasets/
Mr66/itantra-packs](https://huggingface.co/datasets/Mr66/itantra-packs)